Adversarial Domain Adaptation Network With Calibrated Prototype and Dynamic Instance Convolution for Hyperspectral Image Classification

Yi Huang, Jiangtao Peng, *Senior Member, IEEE*, Genwei Zhang, Weiwei Sun, *Senior Member, IEEE*, Na Chen, and Qian Du, *Fellow, IEEE*

***Abstract*— Recently, the adversarial domain adaptation (ADA) methods have been widely investigated and applied in cross-domain hyperspectral image (HSI) classification. However, most ADA algorithms aim to align the cross-domain distribution without focusing on the class separability of the aligned target features and the information of samples within the domain. To address these issues, a new ADA framework based on calibrated prototype and dynamic instance convolution (CPDIC) is proposed in this article for cross-domain HSI classification. The CPDIC is composed of a generator, a calibrated discriminator (CD), and a classifier. The generator includes a static 3-D convolutional network (SCN) and a dynamic instance convolutional network (DICN), where the SCN is used to extract coarse-grained features of HSI and the DICN can extract sample-specific fine-grained features using instance convolutions generated from dynamic instance convolution kernel generation (DCKG) module. As for the generator, the static and dynamic interactive feature extraction network extracts robust domain-invariant features with discriminability. The CD aligns the marginal distribution between domains and calibrate the predicted pseudo-labels of target domain. For classification, a calibrated prototype loss (CPL) is introduced to align the class distribution across domains. The results of three cross-domain HSI classification tasks show that the proposed CPDIC outperforms existing unsupervised domain adaptation (UDA) algorithms.**

***Index Terms*— Adversarial learning, calibrated prototype, domain adaptation (DA), dynamic instance convolution, hyperspectral image (HSI).**

NOMENCLATURE

X_s, X_t	Samples of the source and target domains.
Y_s	Labels of the source domain.
n_s, n_t	Number of samples in the source and target domains.
$P_s(X_s)$, $P_t(X_t)$	Marginal distribution of source and target.
$Q_s(y_s\|X_s)$, $Q_t(y_t\|X_t)$	Conditional distribution of source and target.
G	Feature extractor (i.e., generator).
D	Calibrated discriminator.
F	Classifier.
f	Output features of generator.
f_{SCN}	Features of static 3-D convolutional network.
f_{DICN}	Features of dynamic instance convolutional network.
C	Number of categories.
K_s, K_t	Prototypes of the source and target domains.
$p_s(x)$, $p_t(x)$	Sample distribution of source and target.
$\mathbf{p}_t$	Classifier outputs probability vector of target sample.
L_{adv}	Calibrated discriminator's adversarial loss.
L_{trans}	Transfer loss.
L_c	Calibrated loss.
L_s	Cross-entropy loss on the source samples.
L_{CPL}	Calibrated prototype loss.
$\xi^{(x_t)}$	Calibrated discriminator output vector.
$\eta^{(x_t)}$	Estimated confusion matrix of target sample.
$\hat{y}_t$	Pseudo-label of target sample.
$\tilde{y}_t$	Calibrated pseudo-label of target sample.

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I. INTRODUCTION

HYPERSPECTRAL image (HSI) classification is one of the popular research topics in remote sensing, which can be widely applied in many fields [1], such as agriculture, ecology, environmental resources, military, and applications.

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Recently, many traditional machine learning algorithms have been employed for HSI classification [2], [3]. With the development of deep learning, many HSI classification algorithms based on deep learning have been adopted and achieved remarkable performances [4], [5], [6], [7], [8], [9], [10], [11]. The robust performance of traditional and deep learning-based algorithms mostly relies on a large number of labeled samples. However, the label acquisition of HSIs is usually difficult. For real applications, we often face the situations that some existing historical images are labeled while newly acquired images are unlabeled. The labeled historical image is source domain and the unlabeled new image is target domain. If we use a model trained on source domain to predict target domain, the performance of the model is poor. This is because different HSIs usually have large distribution differences owing to the differences in imaging sensors, imaged geographic areas, acquisition environmental conditions, etc. To alleviate distribution differences between images and adapt a trained model to new images, domain adaptation (DA) technique is proposed to predict target-domain samples using a trained model on labeled source domain. DA has shown good performance by reducing the differences between domains when the target domain has no labels or only a few labels [12]. Among the DA framework, the unsupervised DA (UDA) problem with no labels in target domain is a challenging one and widely investigated.

Recently, in order to solve the cross-domain classification problem, scholars have proposed many UDA methods which achieved good performance [13], [14], [15], [16], [17], [18], [19], [20]. Traditional DA methods can be roughly categorized as instance-based [21], [22], [23], feature-based [16], [24], [25], [26], [27], [28], [29], and classifier-based [30], [31] methods. Most of these methods focus on handcrafted features. To automatically learn data-dependent features, deep DA methods have been proposed, such as discrepancy-based methods [32], [33], [34], [35], [36] and adversarial-based methods [19], [37], [38], [39], [40], [41]. By means of an adversarial manner between a generator and a discriminator, the adversarial-based methods can finally learn domain-invariant features, which is naturally suitable for DA tasks. Saito et al. [39] proposed a maximum classifier discrepancy (MCD) method, which uses dual classifier adversarial training to obtain decision boundaries of domains to drive the alignment of class-level distribution between domains. Xiao and Zhang [40] proposed a dynamic weighted learning (DWL) method, which dynamically weights domain alignment loss and class discrimination loss to ensure that domain-invariant discriminative features are learned. Chen et al. [42] proposed an adversarial-learned loss for DA (ALDA), which combines adversarial learning and self-training to reduce differences between domains and match distribution of feature through a confusion matrix learned in an adversarial manner.

Recently, DA methods have been rapidly developed in the area of HSI classification [13]. In order to learn the subspaces of domains and preserve the discriminability of features, a discriminative cooperative alignment (DCA) method [43] has been proposed to consider both geometrical and statistical alignments. Fang et al. [44] combined DA and confident learning (CL) and proposed a CL-based DA method (CLDA), where DA reduces the domain shift and produces pseudo-label of target samples, and CL improves the network's discriminative ability by selecting high-confidence target samples [44]. Ning et al. [18] proposed a contrastive learning based on category matching (CLCM) method, which performs category matching across domains to reduce the domain shift. Zou et al. [45] consider adapting image-level as well as instance-level features, and introducing accurate and stable target-domain supervision to reduce the distribution shift between domains in the feature space.

The above DA algorithms aim to extract domain-invariant features with discriminable information by matching cross-domain distribution or adversarial training and have achieved good performance on cross-domain HSI classification problem [18], [44], [45], [46]. However, they also have some limitations. An HSI often have complex ground objects and each sample may have different spatial structures. Most current deep networks learn parameters autonomously to extract features, which may lose specific characteristics of the sample, resulting in weak discriminability of features. In addition, existing DA methods intended to simultaneously align marginal and class conditional probability distributions between domains. However, in UDA, the target's labels are inaccessible so that accurate estimation of its class conditional probability distribution is difficult. Many DA methods use self-training strategy to estimate the class distribution of target domain, but their performance heavily depends on the quality of pseudo-labels [42], [47]. Furthermore, for UDA, it should focus not only on the cross-domain distribution alignment but also on the separability of the target features after alignment.

To solve the above problems, a novel adversarial DA (ADA) network based on calibrated prototype and dynamic instance convolution (CPDIC) is proposed. In CPDIC, a dynamic instance convolution network (DICN) is designed to extract sample-specific fine-grained features, and is further combined with an SCN to extract complementary HSI features. In addition, the pseudo-labels of target domain are calibrated by a calibrated discriminator (CD), and a prototype learning strategy is used to align the class-level feature distribution across domains. The contributions of the proposed CPDIC can be summarized as below.

1) The proposed CPDIC that combines adversarial training and self-training in a framework that contains a generator, a discriminator, and a classifier, where the generator is a static and dynamic interactive feature extraction network and the discriminator is a CD. The generator and discriminator are adversarially trained to extract domain-invariant features.
2) The static and dynamic interactive feature extraction network consists of an SCN and a DICN. The coarse-grained features of HSI are extracted by the SCN, and DICN is designed to obtain finer-grained information using a dynamic instance convolution kernel generation (DCGK) module. By performing DCGK on samples, instance convolutions can extract sample-specific fine-grained features. The static–dynamic

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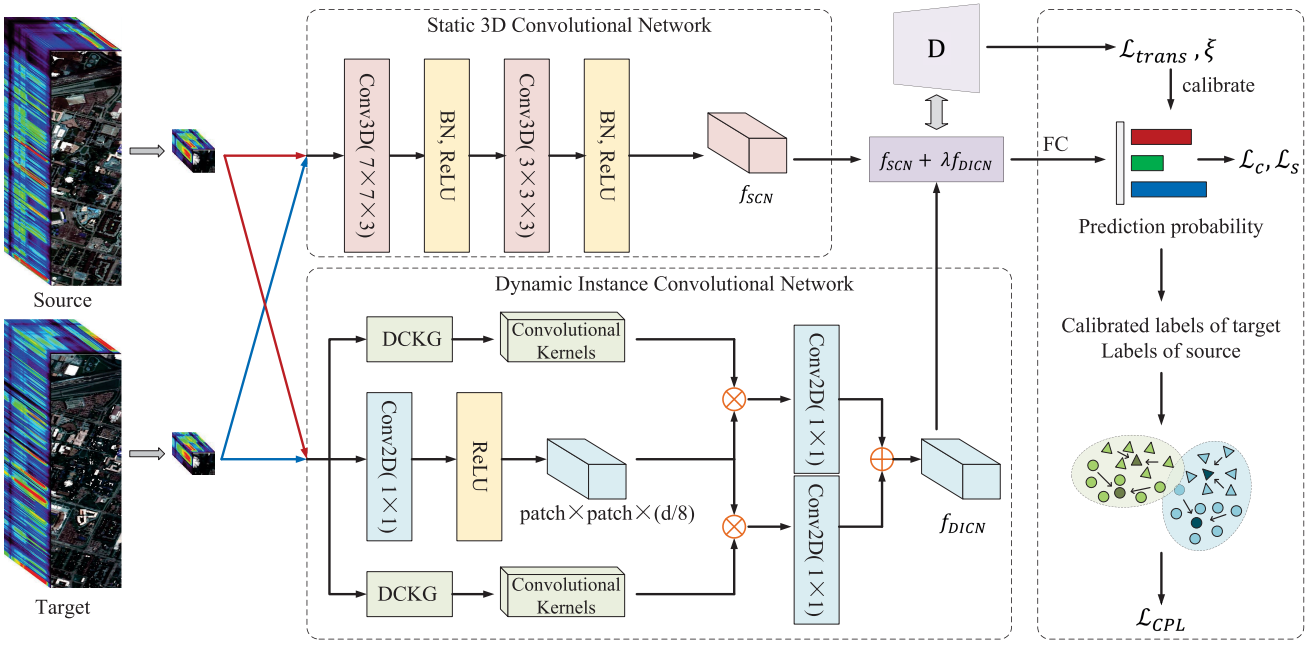

Fig. 1. Framework of the proposed CPDIC. CPDIC includes two main modules: a generator and a calibration discriminator. The generator is composed of an SCN and a DICN, which are used to extract domain-invariant features from HSIs. The CD is used to identify the domain attribution of the features and outputs a calibration vector for calibrating the pseudo-labels of the target domain.

interaction can extract robust domain-invariant features with distinguishability.

3) A CD is designed to adjust the distribution between domains and calibrate the pseudo-labels of the target-domain outputs by the classifier. Meanwhile, a CPL is introduced to align the class-level feature distribution across domains.

The rest of the article is organized as follows. Section II presents the problem and definition, and Section III describes the proposed CPDIC method in detail. The experimental results on three DA tasks are analyzed and discussed in Section IV. Finally, Section V concludes this article.

II. Problem and Definition

Let $X_s = \{x_i^s\}_{i=1}^{n_s}$ denote the n_s samples from the source domain and with labels $Y_s = \{y_i^s\}_{i=1}^{n_s}$, and $X_t = \{x_j^t\}_{j=1}^{n_t}$ be the n_t samples from the target domain without labels. The marginal and class-conditional probability distributions of the source and target domains are inconsistent, i.e., $P_s(X_s) \neq P_t(X_t)$ and $Q_s(y_s|X_s) \neq Q_t(y_t|X_t)$. DA enables the model trained on the source domain to have robust performance on the target domain by aligning the distribution between domains. The definitions of the symbols mentioned in this article are summarized in Nomenclature.

III. Proposed Method

The flowchart of the proposed CPDIC is shown in Fig. 1. The generator G consists of a DICN and an SCN for extracting the HSI features. A CD D is designed to adjust the distribution between domains and calibrate the pseudo-labels of target domain. Based on the calibrated pseudo-labels, the network can introduce the CPL to match the class conditional probability distributions between source and target as shown in Fig. 2.

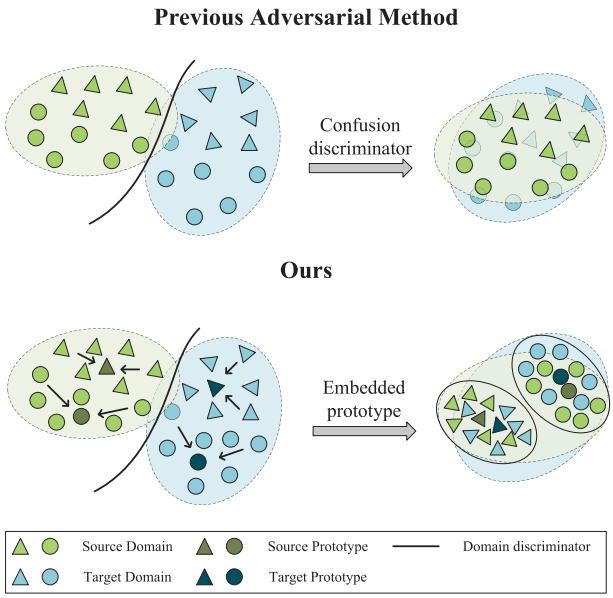

Fig. 2. Top: Existing domain discriminator-based methods can align the global distributions of source and target domains, but lose the class discriminability. Bottom: Our adversarial approach with embedded domain prototype learning learns domain-invariant features with intra-class compactness and inter-class scattering.

A. Generator: Static and Dynamic Interactive Feature Extraction Network

Different from the previous works [34], [39], [48], this article designs a static–dynamic interactive feature extraction network consisting of an SCN and a DICN as shown in Fig. 3. The SCN is used to extract coarse-grained features of HSI. The DCKG module is introduced into the DICN, which can generate instance convolutions based on samples to extract sample-specific

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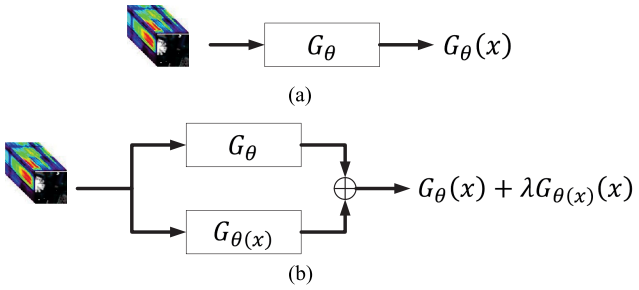

Fig. 3. (a) Previous works aim to use G_θ to automatically learn parameters θ to extract features for all samples consistently. (b) Our work designs a $G_{\theta(x)}$ for each sample x. $G_{\theta(x)}$ generates the parameters $\theta(x)$ based on the sample x and performs on the sample to extract the sample-specific fine-grained features. The parameter λ is used to balance $G_\theta(x)$ and $G_{\theta(x)}(x)$.

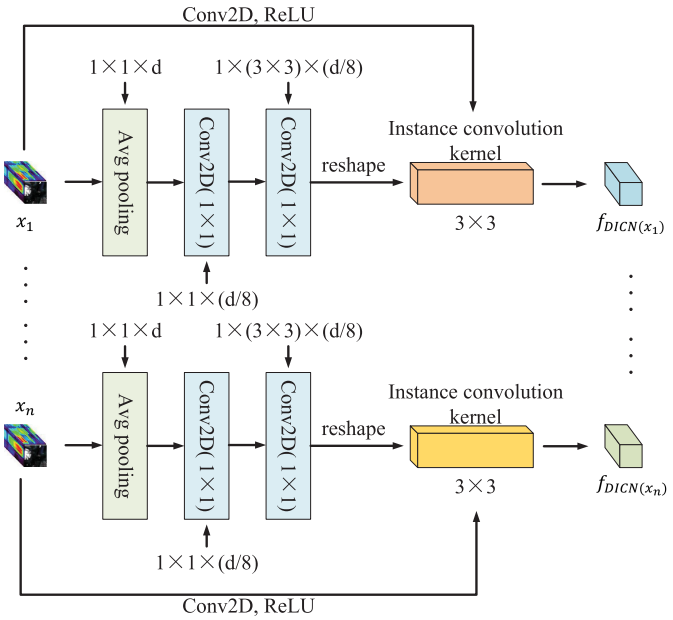

Fig. 4. Specific operation of DCKG acts on different samples.

fine-grained features. As a generator, the static and dynamic interactive feature extraction network can extract robust feature representations.

1) Static 3-D Convolutional Network (SCN): The SCN consists of two layers of 3-D convolution, each followed by a batch normalization (BN) [49] layer and a ReLU activation layer. As shown in Fig. 1, samples of size patch × patch × d are obtained from the source domain and the target domain, respectively, and put into the SCN. The feature f_{SCN} are output after two layers of 3-D convolution.

2) Dynamic Instance Convolutional Network: DICN consists of three branches, the top and down branches are for the generation of two sets of dynamic convolutions, and the middle branch is used to process the input samples. The two sets of output dynamic convolution kernels are further applied to the output data of the middle branch and associated with a 2-D convolution layer. The dynamic instance features f_{DICN} are obtained by combining them in a concatenated way. The DCKG module is shown in Fig. 4, which can generate instance convolution kernels and obtain sample-specific fine-grained features for different samples $x_1, \ldots, x_n$.

3) Combination of Static and Dynamic Features: The generator G combines the f_{SCN} and f_{DICN} as follows:

$$f = f_{\text{SCN}} + \lambda f_{\text{DICN}} \tag{1}$$

where the parameter λ is the similarity between features f_{SCN} and f_{DICN}. The similarity is measured by the normalized L2 norm as in the following equation:

$$\lambda = \text{BN}\left(\| f_{\text{SCN}} - f_{\text{DICN}} \|_2^2\right) \tag{2}$$

where BN$(\cdot)$ is the normalization layer.

Finally, the output features of the static and dynamic interactive feature extraction network are f.

B. Calibrated Prototype Loss

General adversarial DA algorithms can align cross-domain distributions to a certain extent, but they cannot guarantee that the features of the target domain can be distinguished by the classifier F after alignment. In this article, we combine self-training and adversarial learning strategies, and then embed prototype learning to guide the network's adaptive cross-domain class distribution matching.

1) Calibrated Discriminator: For the UDA problem, the prototype of the source can be directly calculated form real labels, but the target domain's labels are unknown. In order to learn high-quality pseudo-labels of the target domain, a CD D and calibrated loss L_c are designed. Among them, the former can not only distinguish domain features, but also calibrate the pseudo-labels of the target domain so that the class feature distributions of target and source are opposite. The latter is a self-training loss based on target-domain samples, which is used to ensure the reliability of pseudo-labels.

Previous works used a confusion matrix to estimate noisy labels [42], [50], [51]. Based on their research, we find that the general self-training loss function is different from the loss function estimated using a confusion matrix. Specifically, given a target sample $x_t \in X_t$ and y_t as its true label, the self-training loss of the target domain under ideal conditions is

$$\mathcal{L}(x_t) = \sum_{i=1}^{C} p(y_t = i | x_t) \mathcal{L}(\mathbf{p}_t, i) \tag{3}$$

where C is the number of categories, and $\mathbf{p}_t$ is the predicted probability vector of x_t output by F and $\mathcal{L}(\mathbf{p}_t, i)$ is the basic loss (e.g., cross-entropy loss function). In UDA, y_t is unknown. Equation (3) can be calculated by using the pseudo label $\hat{y}_t$ of x_t

$$\begin{aligned}
\mathcal{L}(x_t) &= \sum_{i=1}^{C} p(y_t = i | x_t) \mathcal{L}(\mathbf{p}_t, i) \\
&= \sum_{i=1}^{C} \sum_{j=1}^{C} p(y_t = i | \hat{y}_t = j, x_t) p(\hat{y}_t = j | x_t) \mathcal{L}(\mathbf{p}_t, i) \\
&= \sum_{i=1}^{C} \sum_{j=1}^{C} \eta_{ij}^{(x_t)} p(\hat{y}_t = j | x_t) \mathcal{L}(\mathbf{p}_t, i) \\
&= \sum_{i=1}^{C} \tilde{y}_t^i \mathcal{L}(\mathbf{p}_t, i).
\end{aligned} \tag{4}$$

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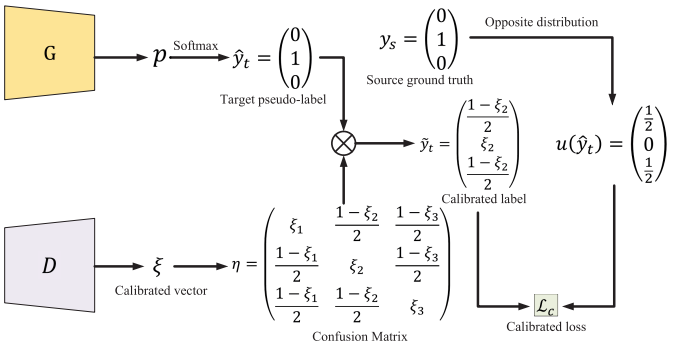

Fig. 5. Specific process of calibrating the target-domain pseudo-labels: the source domain distribution is known and D outputs a calibrated vector $\xi^{(x_t)}$ to operate on the target-domain pseudo-label $\hat{y}_t$, thus directing the target-domain class feature distribution to the opposite distribution of source.

Here, $\eta_{ij}^{(x_t)}$ is the element of confusion matrix $\eta^{(x_t)}$, and $\tilde{y}_t^i = \sum_{j=1}^{C} \eta_{ij}^{(x_t)} p(\hat{y}_t = j|x_t)$ and $\tilde{y}_t$ is the calibrated pseudo-label of target sample x_t. If we can accurately estimate $\eta^{(x_t)}$, pseudo labels of the target domain can be calibrated. However, the estimation of $\eta^{(x_t)}$ is difficult. To simplify the problem, we follow the assumption in [42] that noise in the labels is class-wisely uniformly distributed with the marginal distribution $\xi^{(x_t)}$ where $\xi^{(x_t)} \in \mathbb{R}^C$, $\eta_{ij}^{(x_t)} = \xi_i^{(x_t)}$ for $i = j$, and $\eta_{ij}^{(x_t)} = (1 - \xi_j^{(x_t)})/(C-1)$ for $i \neq j$. In addition, the unhinged loss $\mathcal{L}(\mathbf{p}_t, i) = \mathcal{L}_{unh}(\mathbf{p_t}, i) = 1 - \mathbf{p}_t^i$ is verified to be robust to uniformly distributed noise [52].

In this article, the CD D is designed to learn the calibrated vector $\xi^{(x_t)}$, and the domain adversarial loss based on D is

$$L_{\text{adv}} = L_{\text{trans}} + L_c \tag{5}$$

where

$$L_{\text{trans}} = -\big(E_{x\sim p_s(x)}\big[\log D(G(x))\big] + E_{x\sim p_t(x)}\big[\log(1 - D(G(x)))\big]\big) \tag{6}$$

$$L_c = \sum_{i=1}^{C} \tilde{y}_t^i \mathcal{L}_{unh}(\mathbf{p}_t, i). \tag{7}$$

Taking $C = 3$ as an example, Fig. 5 shows the specific operation of D.

Finally, G and D adversarial training can produce robust domain-invariant features with class separability. Therefore, the classifier F trained from the source domain can achieve robust classification results in the target domain.

2) Prototype-Calibrating Learning: We refer to the prototypes learned via the D-calibrated pseudo-labels of target domain as the calibrated prototypes. Since labels from the source domain are accessible, the prototypes of the source can be calculated directly by the real labels. Specifically, the prototypes are batch-level computed and updated in training by an exponential moving average (EMA) strategy. The batch update rule for the prototype is as follows:

$$K = \lambda_{\text{ema}} K_i + (1 - \lambda_{\text{ema}}) K_{i+1}, \quad i = 1, \ldots, \text{epoch}$$

where λ_{ema} is generally taken as 0.99 and K_{batch} denotes batch-level prototypes.

3) Prototype Loss and Calibrated Prototype Loss: In order to ensure that G can output features of source and target, which have similar class distributions. The CPL is applied to align the class distributions across domains. As shown in (8), the former term is used to decrease the intra-class distance within the domain. The latter term is used to align clusters of the same category between domains. The batch-level CPL designed in this article is

$$L_{\text{CPL}} = \sum_{c=1}^{C} \sum_{i=1}^{\text{batch}} \left(\left\| K_s^c - f_s^i \right\|_2^2 + \left\| K_t^c - f_t^i \right\|_2^2 \right) + \sum_{c=1}^{C} \left\| K_s^c - K_t^c \right\|_2^2. \tag{8}$$

C. Overall Function and Training Steps

The overall loss of the network includes the classification loss of the source domain L_s, the domain adversarial loss of the CD L_{adv}, and the CPL L_{CPL}. During the training process, there is a game between G and D. Among them, G is used for extracting domain-invariant features and D is used for learning inter-domain transferability, with the overall objective of training being

$$\min_{D} L_s + L_{\text{adv}} + \gamma L_{\text{CPL}} \tag{9}$$

$$\min_{G,C} L_s - L_{\text{adv}} + \gamma L_{\text{CPL}} \tag{10}$$

where $L_s = -E_{(x_s,y_s)\sim(X_s,Y_s)} \sum_{c=1}^{C} \mathbf{1}_{[c=y_s]} \log p(y|x_s)$ and γ is a trade-off parameter.

The main steps of the proposed CPDIC are summarized in Algorithm 1.

IV. Experiments

In this section, we design three cross-domain HSI classification tasks to verify the robustness of the proposed CPDIC. The three tasks include Houston data, Shanghai-Hangzhou data, and Pavia datasets, and the tasks address the problem of domain shift caused by different imaging time and different geographical areas. KNN (without DA) and some traditional and deep DA algorithms are used for comparison. Only the labels from the source domain can be used in all the tasks.

A. Data and DA Task

1) Houston Task: The source domain is Houston2013 data [53] and the target domain is Houston2018 data [54]. The common seven categories are used for this task. The Houston2013 data has 349×1905 pixels and includes 144 spectral bands. The Houston2018 data has 48 spectral bands. The Houston2013 data with 48 spectral bands was selected to correspond to the Houston2018 data. Table I shows the categories and the number of samples corresponding to the categories, and Fig. 6 shows the pseudo-color composite image (PCI) and ground-truth (GT) map.

2) H-S Task: The source domain is Shanghai data and the target domain is Hangzhou data with three categories [55]. Shanghai data has 1600×230 pixels and includes 198 spectral bands. Hangzhou data has 590×230 pixels with

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Algorithm 1 CPDIC

Input: X_s, Y_s and X_t.
Parameter: The trade-off parameter γ.
Output: Generator G and classifier F.
The batches of training is $batch_size = 128$:
1. Data pre-processing: The patch samples from source and target domain are obtained according to $patch_size$.
2. Initialize the source and target domain batch-level prototypes: K_s, K_t.
3. Updating network:
Repeat

1) Updating the calibrated discriminator D by fixing the generator G:

$$\min_{D} L_s + L_{adv} + \gamma L_{CPL} \rightarrow D$$

2) Updating the generator G and the classifier F by fixing the calibrated discriminator D:

$$\min_{G,F} L_s - L_{adv} + \gamma L_{CPL} \rightarrow G, F$$

Until convergence
4. Obtain the optimal G and F.
Testing: Fixing generator G and classifier F, input target domain samples X_t and output prediction labels:

$$Y_t = argmax(F(G(X_t)))$$

TABLE I
NUMBER OF SAMPLES IN THE HOUSTON TASK

Houston13 and Houston18			
No.	Class	Source (Houston13)	Target (Houston18)
1	Grass healthy	345	1353
2	Grass stressed	365	4888
3	Trees	365	2766
4	Water	285	22
5	Residential buildings	319	5347
6	Non-residential buildings	408	32459
7	Road	443	6365
Total		2530	53200

TABLE II
NUMBER OF SAMPLES IN THE H-S TASK

Hangzhou and Shanghai			
No.	Class	Source (Hangzhou)	Target (Shanghai)
1	Water	18043	123123
2	Land / building	77450	161689
3	Plant	40207	83188
Total		135700	368000

198 spectral bands. Table II shows the categories and the number of samples corresponding to the categories, and Fig. 7 shows the PCI and GT map.

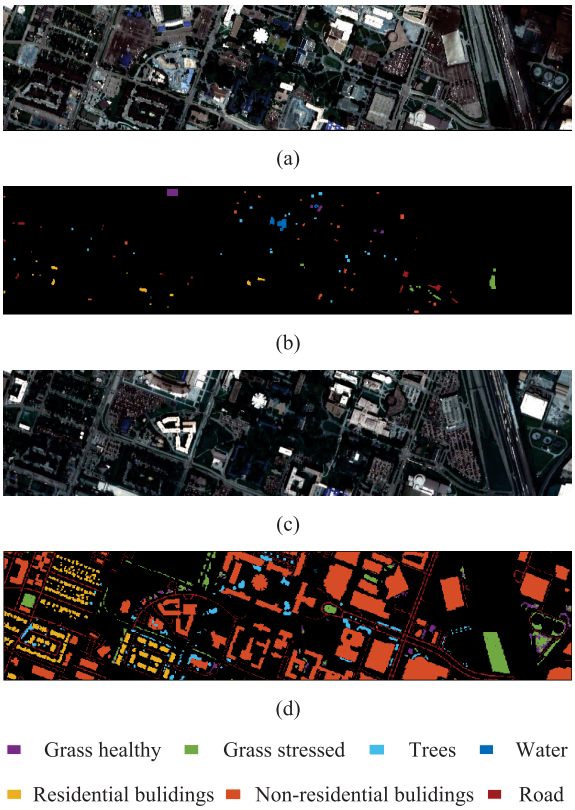

Fig. 6. Houston task. (a) PCI of Houston13. (b) GT map of Houston13. (c) PCI of Houston18. (d) GT map of Houston18.

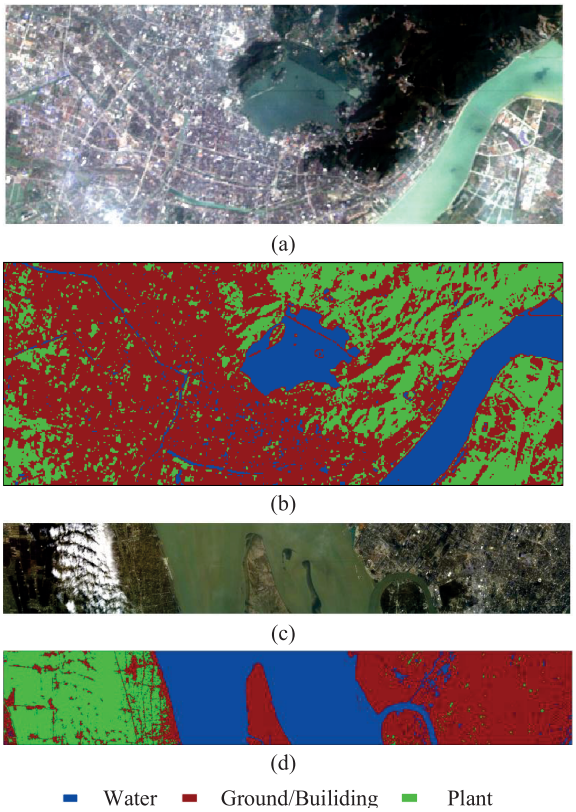

Fig. 7. H-S task. (a) PCI of Hangzhou. (b) GT map of Hangzhou. (c) PCI of Shanghai. (d) GT map of Shanghai.

3) Pavia Task: The Pavia datasets contain the Pavia University (UP) and the Pavia Center (PC) in Pavia, Northern Italy,

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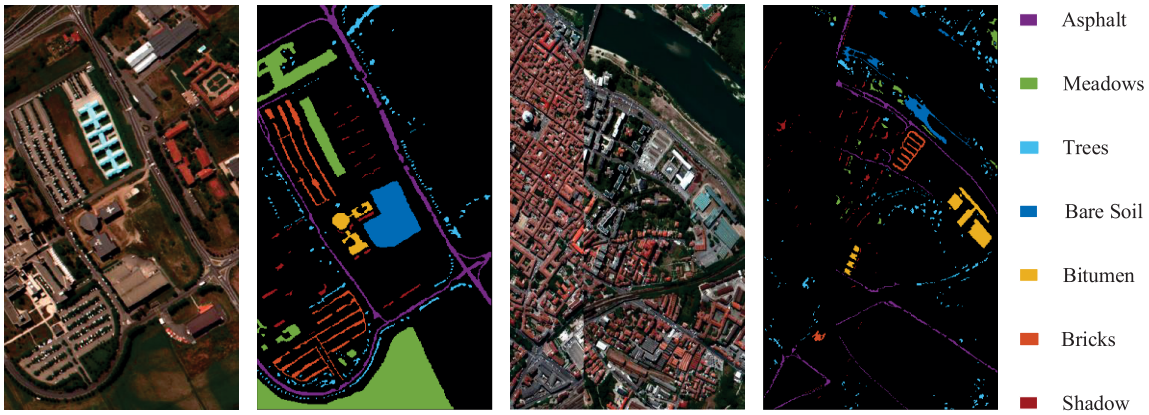

Fig. 8. Pavia task. (a) PCI of UP. (b) GT map of UP. (c) PCI of PC. (d) GT map of PC.

TABLE III
NUMBER OF SAMPLES IN THE PAVIA TASK

Pavia University and Pavia Center			
No.	Class	Source (UP)	Target (PC)
1	Asphalt	6631	9248
2	Meadows	18649	3090
3	Trees	3064	7598
4	Bare soil	5029	6584
5	Bitumen	1330	7287
6	Bricks	3682	2685
7	Shadow	947	2863
Total		39332	39355

both collected by ROSIS sensors. The UP serves as the source domain with 610 × 340 pixels and 103 bands, the PC serves as the target domain with 1096 × 715 pixels and 102 bands. The common seven categories are used in the classification task. Table III shows the categories and the number of samples corresponding to the categories, and Fig. 8 shows the PCI and GT map.

B. Experiment Setting and Parameter Tuning

Three cross-domain HSI classification tasks are used to validate the robustness of the proposed CPDIC. Traditional DA methods: KNN, SA [24] and deep DA methods: DAN [32], DAAN [38], DSAN [34], MRAN [35], ADDA [37], DIDA [23], TSTnet [48] are used for comparison. 20 is set as the dimension of the subspace of SA, the base learning rate lr of the DA algorithms based on deep learning is selected from $\{0.0001, 0.001, 0.01, 0.1\}$, and the training batch size and the number of iterations are uniformly set to 128 and 100, respectively. In addition, DAN, DAAN, DSAN, and MRAN have a regularization parameter, and this parameter is selected from $\{0.001, 0.01, 0.1, 1, 10, 100\}$. In addition, the optimal hyperparameters of all algorithms in the experiment are selected based on the optimal value search range proposed in the source paper. All experiments are performed under the Pytorch, and the base network is the ResNet18 [56]. In the proposed CPDIC, the optimal learning rate is 0.01 and the parameter γ in the three DA tasks is set to 0.1.

Prototype learning is embedded in CPDIC, so it can achieve robust results using only a small number of source-domain labels. In the setting of the DA task, except for TSTnet and the proposed CPDIC, all comparison algorithms use 80% of the source samples and labels. For TSTnet and CPDIC, only 5% of the source samples and labels are used.

C. Results and Performance on Cross-Scene HSI Classification

Tables IV–VI show the classification results of the comparison algorithms and the proposed CPDIC for the three cross-domain HSIs classification tasks, where the classification performance is evaluated in terms of overall accuracy (OA) and kappa coefficient (κ). According to Tables IV–VI, the analysis is as follows.

1) The algorithm with the best overall performance among non-adversarial DA algorithms is DIDA. Among the adversarial DA algorithms, ADDA has a better performance in the Pavia task, and DAAN has a better performance in the Houston task and the H-S task. However, the performance of the adversarial DA algorithms is lower than DIDA, which shows that it is necessary to consider sample-specific fine-grained features when extracting features of HSIs. DCKG is designed in CPDIC for extracting sample-based fine-grained features, while CD D is designed to learn target-domain features with class separability. The classification performance of the proposed CPDIC outperforms DIDA in the three DA tasks.
2) The performance of TSTnet in the Houston task and the H-S task is better than other compared DA algorithms, which shows that cross-domain HSI classification problem needs to consider the characteristics of HSI. In CPDIC, we use static and dynamic interactive feature extraction network to extract the spatial–spectral joint features of HSIs and design a CPL L_{CPL} to ensure the alignment of cross-domain class distribution, which outperforms TSTnet in three DA tasks.

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TABLE IV
CLASSIFICATION ACCURACIES FOR THE HOUSTON TASK (TSTNET AND CPDIC USE 5% OF THE LABELED SOURCE SAMPLES)

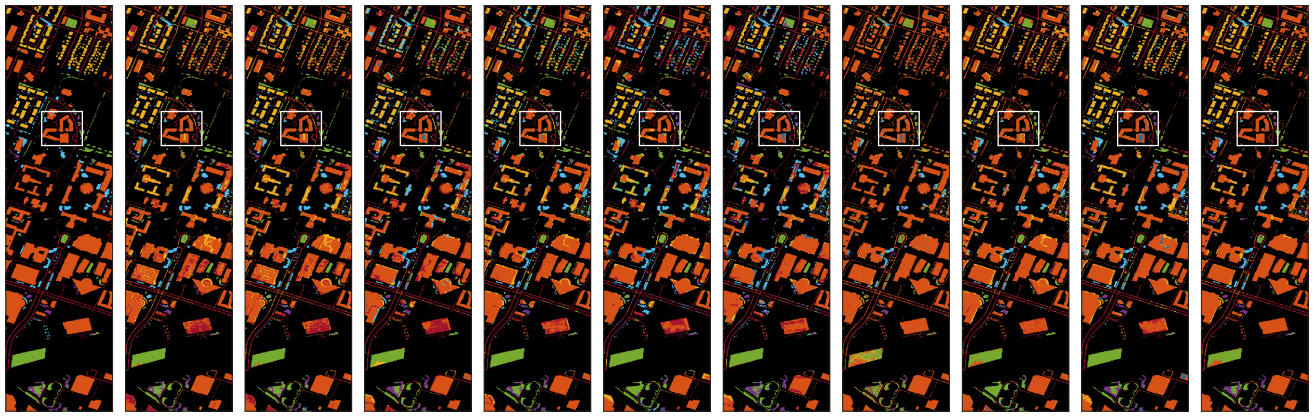
Class	Classification algorithms									
	KNN	SA	DAN	DAAN	DSAN	MRAN	ADDA	DIDA	TSTnet	CPDIC
1	78.94	78.79	63.78	79.66	26.46	58.31	54.84	45.00	**85.03**	41.11±0.07
2	**94.82**	**94.82**	75.57	74.09	78.07	74.00	74.63	76.73	68.73	90.33±0.03
3	66.52	66.56	62.80	65.00	**73.17**	67.53	63.56	60.70	52.82	57.68±0.11
4	**100**	**100**	**100**	**100**	**100**	81.82	**100**	**100**	**100**	80.45±0.08
5	64.00	64.02	46.04	59.12	41.05	55.71	58.01	**65.36**	61.71	46.19±0.47
6	68.10	68.23	69.89	76.01	72.16	69.07	73.87	76.86	84.44	**88.02±0.08**
7	60.91	60.71	**73.44**	54.95	61.30	58.55	62.91	59.05	53.07	65.09±0.41
OA(%)	69.49	69.55	67.92	71.18	67.18	66.57	70.03	71.92	75.34	**78.51±0.07**
KC×100(κ)	53.86	53.92	52.06	55.33	50.76	50.91	52.40	55.22	59.69	**63.90±0.13**

TABLE V
CLASSIFICATION ACCURACIES FOR THE H-S TASK (TSTNET AND CPDIC USE 5% OF THE LABELED SOURCE SAMPLES)

Class	Classification algorithms									
	KNN	SA	DAN	DAAN	DSAN	MRAN	ADDA	DIDA	TSTnet	CPDIC
1	93.35	91.43	86.88	88.56	93.48	85.69	85.88	92.61	93.47	**97.19±0.03**
2	41.51	63.74	82.27	70.28	82.45	84.27	73.84	81.75	84.58	**95.98±0.08**
3	**100.00**	95.32	99.20	99.95	99.49	98.38	87.71	99.25	96.96	92.69±0.24
OA(%)	72.07	80.14	87.64	82.98	89.99	87.93	81.00	89.34	90.36	**95.64±0.78**
KC×100(κ)	59.90	70.48	81.14	74.70	84.79	81.52	71.16	83.80	85.27	**93.21±1.89**

TABLE VI
CLASSIFICATION ACCURACIES FOR THE PAVIA TASK (TSTNET AND CPDIC USE 5% OF THE LABELED SOURCE SAMPLES)

Class	Classification algorithms									
	KNN	SA	DAN	DAAN	DSAN	MRAN	ADDA	DIDA	TSTnet	CPDIC
1	77.10	**98.86**	84.49	80.98	85.39	82.78	86.28	88.40	91.96	90.37±0.02
2	87.48	80.00	88.91	88.90	79.92	**90.45**	37.90	82.01	90.87	73.29±0.28
3	85.17	87.87	75.60	71.16	89.34	70.05	82.84	84.59	55.43	**93.03±0.05**
4	68.67	73.07	86.40	80.14	73.15	72.66	**90.73**	77.00	81.77	81.51±0.06
5	1.51	0.77	71.95	67.08	67.16	64.43	69.72	73.93	62.47	**79.71±0.08**
6	33.78	49.42	66.01	68.58	74.90	72.42	17.19	67.31	53.50	**87.68±0.07**
7	83.67	**99.93**	77.02	74.89	86.86	74.45	93.68	74.17	84.72	80.40±0.26
OA(%)	61.61	69.49	79.31	75.70	79.65	74.53	77.25	80.10	74.51	**85.18±0.22**
KC×100(κ)	53.59	62.87	75.27	75.56	77.09	69.77	72.41	75.17	69.59	**82.19±0.30**



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k)

Fig. 9. Houston task. (a) GT. (b) KNN (69.49%). (c) SA (69.55%). (d) DAN (67.92%). (e) DAAN (67.92%). (f) DSAN (67.18%). (g) MRAN (66.57%). (h) ADDA (70.03%). (i) DIDA (71.92%). (j) TSTnet (75.34%). (k) CPDIC (78.51%). (The white boxes in the figures are the locations where CPDIC is classified more accurately than the comparison algorithms.)

The classification maps are shown in Figs. 9–11. By comparing with GT, it is obvious that CPDIC classifies better than other comparison algorithms. In Figs. 9–11, the white boxes circle the regions where CPDIC classifies more correctly than comparison algorithms. Specifically, the circled regions correspond to the orange class ("non-residential buildings") in

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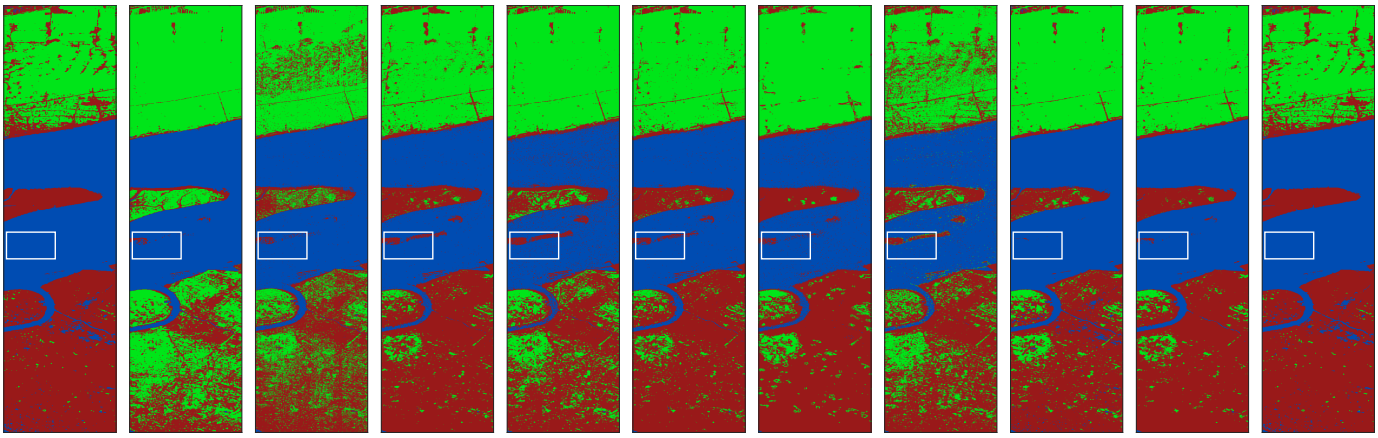
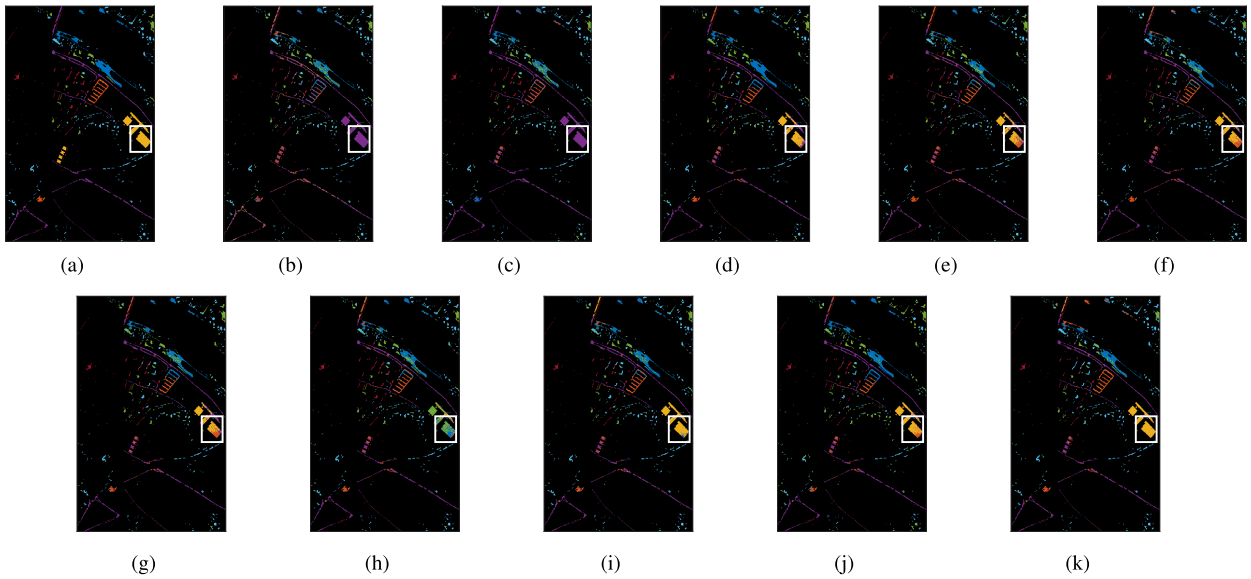


(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k)

Fig. 10. H-S task. (a) GT. (b) KNN (72.07%). (c) SA (80.14%). (d) DAN (87.64%). (e) DAAN (82.98%). (f) DSAN (89.99%). (g) MRAN (87.93%). (h) ADDA (81.00%). (i) DIDA (89.34%). (j) TSTnet (90.36%). (k) CPDIC (95.64%). (The white boxes in the figures are the locations where CPDIC is classified more accurately than the comparison algorithms.)



(a) (b) (c) (d) (e) (f)

(g) (h) (i) (j) (k)

Fig. 11. Pavia task. (a) GT. (b) KNN (61.61%). (c) SA (69.49%). (d) DAN (79.31%). (e) DAAN (75.70%). (f) DSAN (79.65%). (g) MRAN (74.53%). (h) ADDA (77.25%). (i) DIDA (80.51%). (j) TSTnet (74.51%). (k) CPDIC (85.18%). (The white boxes in the figures are the locations where CPDIC is classified more accurately than the comparison algorithms.)

TABLE VII
ABLATION STUDY ON THE THREE TASKS

Task	ResNet18	No SCN	No DICN	No CPL	No CD	CPDIC
Houston task	73.81	72.54	76.06	76.42	76.88	**78.51**
H-S task	87.84	94.46	93.52	94.22	93.10	**95.64**
Pavia task	68.24	82.74	83.41	82.92	80.62	**85.18**

TABLE VIII
OA OF CPDIC FOR DIFFERENT PATCH SIZES

Task	3×3	5×5	7×7	9×9
Houston task	74.39	75.03	77.67	**78.51**
H-S task	85.51	**95.64**	95.49	94.54
Pavia task	83.33	**85.18**	82.91	82.54

the Houston task, the blue class ("Water") in the H-S task, and the yellow class ("Bitumen") in the Pavia task, respectively.

D. Ablation Analysis and Discussion

1) Ablation Analysis: This section analyzes the importance of each module in the proposed CPDIC. The main modules of CPDIC are SCN, DICN, CD, and CPL. In addition, ResNet18 [56] refers to the result of directly predicting the target domain after training using all labeled samples of the source domain.

To verify their effectiveness, we remove each module while fixing others. It can be seen from Table VII that the performance of CPDIC decreases after removing each of the four modules. In particular, without CD, the performance on the Pavia task drops by 4.56%, which shows that it is meaningful to maintain the separability of target-domain features based on adversarial learning. In addition, when DICN is deleted, the performance on the H-S task drops by 2.12%. This shows

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TABLE IX
EXECUTION TIME (IN SECONDS) FOR ONE EPOCH TRAINING OF DIFFERENT METHODS (TSTNET AND CPDIC USE 5% OF THE LABELED SOURCE SAMPLES)

Task	DAN	DAAN	DSAN	MRAN	ADDA	DIDA	TSTnet	CPDIC
Houston task	1.16	1.31	1.47	1.21	1.58	5.25	2.01	2.36
H-S task	119.36	113.77	124.53	121.39	166.14	271.38	28.86	26.19
Pavia task	26.56	28.41	28.47	27.84	31.75	70.98	7.21	8.92

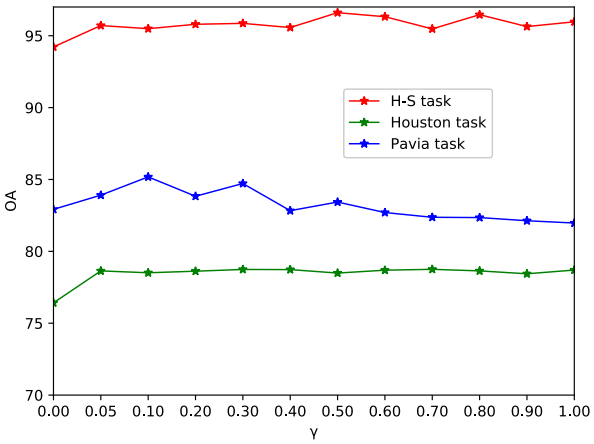

Fig. 12. Influence of parameter γ on OA in the three tasks.

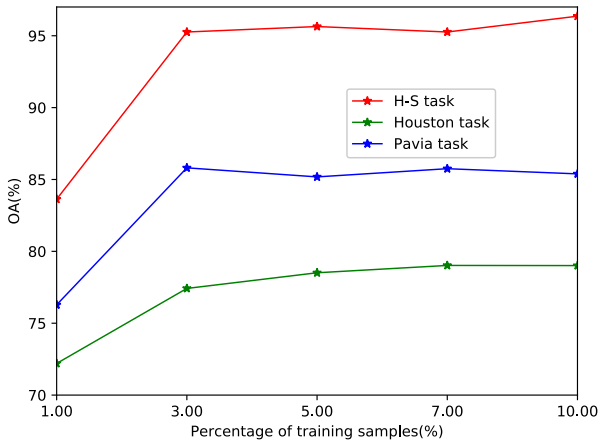

Fig. 13. Overall classification accuracy of CPDIC versus different percentages of training samples for the three tasks.

that considering sample-specific characteristics can assist the network to obtain more robust features with discriminative information.

2) Parameter Analysis: To analyze the impact of the hyperparameter γ on the performance of CPDIC in the three tasks, Fig. 12 shows the OA of CPDIC in three tasks during the change of γ, where $\gamma = 0$ means that L_{CPL} is not used. It can be seen that the performance of CPDIC keeps stable with the change of parameter γ. In the experiments, γ is set as 0.1.

Further, we analyzed the impact of patch size on the performance of algorithm. Table VIII provides the OA of CPDIC in the three tasks when the patch size is taken as 3×3, 5×5, 7×7, and 9×9. According to the results, the patch size takes 9×9 in the Houston task and the patch size takes 5×5 in the Pavia and H-S tasks.

Fig. 13 illustrates the OA of all methods with different percentages of training samples. The proportion of labeled samples in the source domain for the three tasks all ranges from 1% to 10%. The results show that as the proportion of labeled samples in training increases, the prediction accuracy of CPDIC gradually increases. Based on the performance and computational efficiency of the algorithm, we select 5% of the labeled samples in the source domain as training samples. Among them, when the training samples are only 1% of the labeled samples in the source domain, the performance of the algorithm is poor. This is because there are insufficient labeled samples for training and the network cannot learn the accurate data distribution.

3) Feature Visualization: To visualize the performance of CPDIC, we show the TSNE figures of the original data with features being extracted by ADDA and CPDIC on the three DA tasks in Fig. 14. It can be observed that the proposed CPDIC can match the marginal distributions of the source and target domains as well as the class conditional probability distributions to some extent.

In the Houston task, the categories of the original data are confused, and the distribution of samples of each category between domains is inconsistent. Comparing Fig. 14(a) and (b), we find that ADDA can cluster various sample features, but it cannot align the distributions of the same category between domains. CPDIC uses CPL to guide the network to align the class conditional probability distribution across domains, as shown in Fig. 14(c). According to Fig. 14(d), it can be seen that the green category of the target domain is confused with the red category of the source domain in the H-S task. The ADDA can roughly separate and align the three categories between domains. However, there is still confusion between some categories of the target domain as shown in Fig. 14(e). As shown in Fig. 14(f), CPDIC can separate the red and green categories of the source and target domains and align the green categories between domains, so CPDIC can correctly classify the green categories as shown in Table V. In particular, from the comparison of (g)–(i) in Fig. 14, it can be found that CPDIC aligns the blue ("Bitumen") and yellow ("Bricks") categories of the source and target correspondingly, which results in the best prediction performance of the classifier F on these two classes corresponding to Table VI.

4) Computational Complexity Evaluation: In this section, we compare the running time of each deep DA algorithm on

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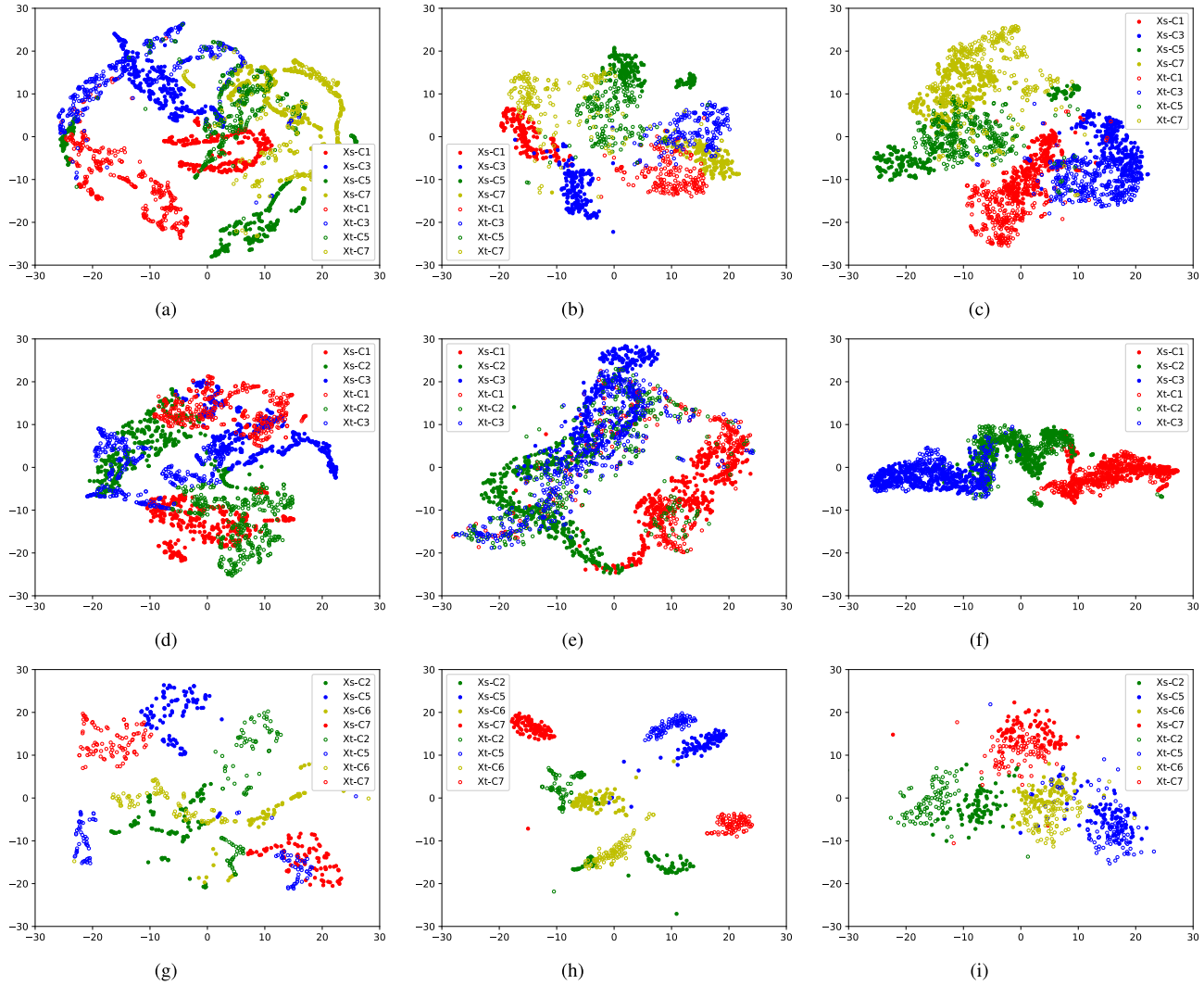

Fig. 14. Cross-domain class distribution alignment performance for three DA tasks. (a) Original features on the Houston task. (b) Features of CPDIC on the Houston task. (c) Features of ADDA on the Houston task. (d) Original features on the H-S task. (e) Features of ADDA on the H-S task. (f) Features of CPDIC on the H-S task. (g) Original features on the Pavia task. (h) Features of ADDA on the Pavia task. (i) Features of CPDIC on the Pavia task.

three tasks. Among the three tasks, the one-epoch running time of all deep DA algorithms is shown in Table IX. All experiments are conducted on a computer based on the Pytorch with a 2.7 GHz Intel Xeon Gold 6258R CPU and four NVIDIA GeForce RTX 2080 Ti GPUs. Under the same experimental settings, CPDIC's running time is slightly higher than TSTnet in Houston and Pavia tasks, but its prediction accuracy is higher than all comparison algorithms in the three tasks.

V. Conclusion

In this article, a new ADA network is proposed, i.e., CPDIC based on calibration prototype and dynamic instance convolution, for cross-domain HSIs classification. The CPDIC is composed of a generator, a CD and a classifier. The generator includes an SCN and a DICN, where the SCN is used to extract coarse-grained features of HSI and the DICN can extract sample-specific fine-grained features using instance convolutions generated from DCKG module. The CPDIC calibrates pseudo-labels of the target domain which is used to learn the calibrated prototype by designing the CD. Finally, it aligns the class distributions between source and target by the CPL. Experimental results demonstrate that the CPDIC is more robust than existing traditional and deep DA methods in the three cross-scene HSIs classification tasks.

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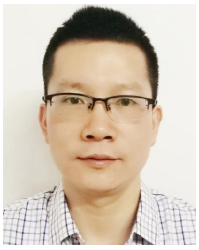

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